

ECON7335: Problem Set 2
“Chunky” vs “Smooth”: the Eternal Battle

1. **Chunky information acquisition.** Consider a static economy in which a continuum of agents try to track an exogenous fundamental θ , as well as the aggregate action in the economy. That is, each agent’s loss function is given by

$$\mathcal{L}^i = E^i [(1 - \alpha)(a_i - \theta)^2 + \alpha(a_i - a)^2] \quad (1)$$

The fundamental in an i.i.d. standard normal, $\theta \sim N(0, 1)$, and agents understand this. Now assume that each agent has the opportunity to “purchase” full information. If they purchase information, their loss is increased by the additive constant c but their actions can be improved by the additional information. The question is: when is information acquisition an equilibrium?

- (a) First, assume that all other agents do not acquire information, and so take the unconditional action $a_j = 0, \forall j \neq i$. Compute the loss function for the individual i assuming i does not acquire the information.
- (b) Now compute the loss function of agent i , if they do acquire the information (maintaining the assumption that other agents do not acquire the information.)
- (c) Comparing welfare in the case above, provide a condition on c such that non-acquisition of information is an equilibrium.
- (d) Now assume that all agents $j \neq i$ do purchase the information at the utility cost c , so that $a_j = \theta, \forall j \neq i$. Compare the welfare to agent i of purchasing and not purchasing the information, to arrive at a condition on c under which information acquisition is an equilibrium.
- (e) Combine your results above to provide a range of value of c in which acquisition and non-acquisition are both equilibria. Does this range get bigger or smaller when α is larger? What happens in the limit as $\alpha \rightarrow 1$, and why?

2. **Smooth information acquisition.** Consider the environment above, with the same fundamental and loss function. The difference is that now, instead of information being all-or-nothing, agents can acquire a noisy signal about the truth,

$$s_i = \theta + \zeta_i, \quad (2)$$

where $\zeta_i \sim N(0, \sigma^2/z_i)$ and z_i is agent i 's chosen precision. The utility cost of the signal is $c(z_i) = \gamma z_i$. In other words, agents can smoothly increase the precision of their signal s_i by paying a higher cost γz_i . Throughout, assume that noise is idiosyncratic, so that $\int \zeta_i di = 0$. It will be convenient to write $\sigma_i^2 \equiv \sigma^2/z_i$ for the signal variance of agent i , and $\hat{\sigma}^2 \equiv \sigma^2/\hat{z}$ for that of the other agents.

- (a) Conjecture that *all* agents, including agent i , use the same fixed information strategy $\hat{z}$, and take optimal solutions for action given that strategy. Find the equilibrium for fixed $\hat{z}$ by conjecturing a relationship $a_i = \beta(\hat{z})s_i$, and solving for $\beta(\hat{z})$.
- (b) Using your solution above, compute the loss of an agent i who chooses a precision z_i as a function of the precision of other agent's information $\hat{z}$. Don't forget to include the information cost γz_i in the loss. (You should arrive at

$$\mathcal{L} = \left(1 - \alpha + \alpha \hat{\beta}^2\right) - \frac{A^2}{1 + \sigma^2/z_i} + \gamma z_i, \quad A \equiv 1 - \alpha + \alpha \hat{\beta}, \quad (3)$$

where $\hat{\beta} \equiv \beta(\hat{z})$ is your answer to (a). Check your algebra against this before moving on — the remaining parts all build on it.)

- (c) Use above loss function to take first-order condition that describes the individually-optimal z .
- (d) Compute the optimal best response $z_i(\hat{z})$ of the agent, given the choice $\hat{z}$ of other agents.
- (e) *Challenge: Find the fixed-point z^* of the best response above as the solution to a quadratic equation in z . Use this to show the economy has a unique linear equilibrium — that is, a unique symmetric equilibrium in strategies of the form $a_i = \beta s_i$.
- (f) Compare your answer in 2.e to your answer in 1.e. Why does the desire to coordinate generate multiple equilibria when information is chunky, but a unique equilibrium when it is smooth?